

Q- Explain in your own words how NAT and PAT help apps like Zomato or Flipkart serve millions of users with limited public IP addresses. Give two real-world scenarios where incorrect NAT/PAT setup could break connectivity for users.

NAT and PAT in real-world apps

Apps like **Zomato and Flipkart** have millions of users, but they cannot give every device its own public IPv4 address because public IP addresses are limited.

NAT (Network Address Translation) solves this by converting private IP addresses into public IP addresses when devices communicate with the internet.

PAT (Port Address Translation) goes one step further. It allows **many devices to share the same public IP address** by using different port numbers to keep their connections separate.

For example:

User 1: 192.168.1.10:5001 └─┐
User 2: 192.168.1.11:5002 └─┼─→ Public IP: 49.x.x.x → Internet
User 3: 192.168.1.12:5003 └─┘

So, thousands of users can access websites and apps while using a much smaller number of public IP addresses. This helps save IPv4 addresses and makes network management easier.

Two real-world problems caused by incorrect NAT/PAT

1. Online payment or login failure

Suppose a user is using Zomato and the payment service requires a connection back to the user's network. If NAT is incorrectly configured or required ports are not translated properly, the connection can fail. The user might see a **payment timeout or failed transaction**.

2. Apps or services become unreachable

Suppose a company configures PAT incorrectly on its router. If the wrong interface is marked as ip nat inside or ip nat outside, users may have internet connectivity problems. For example, Flipkart pages might not load, APIs could time out, or users could be unable to log in.